



# Therapeutic effects against high-grade glioblastoma mediated by engineered induced neural stem cells combined with GD2-specific CAR-NK

Weihua Liu<sup>1,2</sup> · Yu Zhao<sup>1,2</sup> · Zhongfeng Liu<sup>1,2</sup> · Guangji Zhang<sup>1,2</sup> · Huantong Wu<sup>1,2</sup> · Xin Zheng<sup>1,2</sup> · Xihe Tang<sup>3</sup> · Zhiguo Chen<sup>1,2,4</sup>

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## Abstract

**Purpose** High-grade glioblastoma is extremely challenging to treat because of its aggressiveness and resistance to conventional chemo- and radio-therapies. On the contrary, genetic and cellular immunotherapeutic strategies based on the stem and immune cells are emerging as promising treatments against glioblastoma (GBM). We aimed to develop a novel combined immunotherapeutic strategy to improve the treatment efficacy using genetically engineered PBMC-derived induced neural stem cells (iNSCs) expressing HSV-TK and second-generation CAR-NK cells against GBM.

**Methods** iNSCs cells expressing HSV-TK (iNSCs<sup>TK</sup>) and GD2-specific CAR-NK92 (GD2NK92) were generated from PBMC-derived iNSCs and NK92 cell lines, respectively. The anti-tumor effect of iNSCs<sup>TK</sup> and the combinational therapeutics of iNSCs<sup>TK</sup> and GD2NK92 were evaluated by GBM cell line using *in vitro* and *in vivo* experiments.

**Results** PBMC-derived iNSCs<sup>TK</sup> possessed tumor-tropism migration ability *in vitro* and *in vivo*, which exhibited considerable anti-tumor activity via bystander effect in the presence of ganciclovir (GCV). iNSCs<sup>TK</sup>/GCV could slow GBM progression and prolong median survival in tumor-bearing mice. However, the anti-tumor effect was limited to single therapy. Therefore, the combinational therapeutic effect of iNSCs<sup>TK</sup>/GCV and GD2NK92 against GBM was investigated. This approach displayed a more significant anti-tumor effect *in vitro* and in xenograft tumor mice.

**Conclusions** PBMC-derived iNSCs<sup>TK</sup> showed a significant tumor-tropic migration and an effective anti-tumor activity with GCV *in vitro* and *in vivo*. In addition, combined with GD2NK92, iNSCs<sup>TK</sup> therapeutic efficacy improved dramatically to prolong the tumor-bearing animal model's median survival.

**Keywords** High-grade glioblastoma · Induced neural stem cells · GD2-specific chimeric antigen receptor engineered NK cells · Bystander anti-tumor effect · Immunotherapy

## 1 Introduction

Cancers of the central nervous system (CNS) are severe diseases, and GBM is the most common and aggressive primary brain tumor. More than 15,000 patients are diagnosed with GBM every year [1, 2], with a median survival expectancy of < 15 months, and < 10% survival rate beyond five years, despite the application of conventional multimodal treatment containing surgical resection, chemotherapy, and radiation therapy [1, 3, 4]. The efficiency of conventional therapies is limited, due to the lack of specificity and the accuracy to eradicate aggressive diffusing GBM cells.

The herpes simplex virus thymidine kinase (HSV-TK) gene is a suicide gene. It exhibits cytotoxicity by monophosphorylating ganciclovir with high efficiency [5–7]. The

Weihua Liu and Yu Zhao contributed equally to this work.

✉ Zhiguo Chen  
chenzhiguo@gmail.com

<sup>1</sup> Cell Therapy Center, Beijing Institute of Geriatrics, Xuanwu Hospital Capital Medical University, National Clinical Research Center for Geriatric Diseases, and Key Laboratory of Neurodegenerative Diseases, Ministry of Education, Beijing 100053, China

<sup>2</sup> Center of Neural Injury and Repair, Beijing Institute for Brain Disorders, Beijing 100069, China

<sup>3</sup> Neurosurgery Center of Aeronautical General Hospital, Beijing 100012, China

<sup>4</sup> Beijing, China

HSV-TK/GCV system has a bystander effect of eliminating the surrounding tumor cells not transduced with the HSV-TK gene [5, 7, 8]. The safety of the HSV-TK/GCV system against GBM therapy was confirmed in phase III clinical trial by Rainov in 2000 [9]. However, the overall survival of patient was not improved. The viral vector's low efficiency of gene transduction and the absence of targeting might contribute to this failure.

Neural stem cells (NSCs) display unique tumor tropism and can naturally migrate toward and invade GBM deposits in response to chemotactic signals released by the cancer cells [10–12]. NSCs delivery of therapeutic molecules is an emerging approach to achieve effective anti-tumor cytotoxicity for tumor cells not sensitive to conventional treatments [13–15]. However, there are ethical concerns regarding aborted fetus-derived neural stem cells.

The development of induced pluripotent stem cell (iPSCs) technology has provided an alternative strategy for cell therapy against diseases [16, 17]. Neural stem/progenitor cells (NS/PCs) derived from human hiPSCs showed remarkable anti-GBM activity in a pre-clinical model [18]. Additionally, mice and human fibroblasts can also be directly trans-differentiated into induced neural stem cells (iNSCs) [19–22]. However, procuring appropriate cells for routine clinical use is one of the greatest challenges to obtaining enough therapeutic NSCs, considering the high risk of teratoma with iPSCs and the difficulty of obtaining human fibroblasts for trans-differentiation use before surgery [23]. Chen's lab successfully generated therapeutic iNSCs using adult donor PBMCs without passing through an undifferentiated pluripotent state [24]. This strategy did not form teratomas in the nude mice model, and it might serve as a more optimal option for treating patients with GBM. In this study, we successfully generated therapeutic iNSCs expressing HSV-TK (iNSCs<sup>TK</sup>) by genetically modified iNSCs derived from adult donor PBMCs.

It is difficult to eliminate all GBM cells through a single therapy, due to the complicated tumor environment and the high degree of inter- and intra-tumor heterogeneities [25, 26]. Regarding the obstacles above, immunotherapy appears to be a feasible approach demonstrating remarkable improvements in managing several solid tumors, including melanoma, renal cell carcinoma, non-small cell lung cancer, and GBM [1, 27, 28]. The disialoganglioside (GD2) is expressed highly and consistently in patient-derived specimens, and the GD2 on the tumor cell surface was suggested as a promising target for chimeric antigen receptors T cells (CAR-T) in pre-clinical GBM models [29, 30]. Similarly, NK cells can be genetically modified to express a CAR that recognizes tumor-associated cell surface antigens and mediates specific recognition and lysis of cancer cells [31, 32]. A human NK cell line (NK92) derived from a patient with non-Hodgkin's lymphoma has

been safely applied as an allogeneic cell in clinical trials [31]. CAR-NK cells have advantages in safety profile compared with CAR-T cells, including reduced risk of graft-versus-host disease (GvHD), cytokine release syndrome, and neurotoxicity [33]. In this study, NK92 was genetically modified to express GD2-CAR using a lentiviral vector stably, and its anti-tumor ability combined with iNSCs<sup>TK</sup> in the presence of GCV was investigated against GBM.

## 2 Methods and materials

### 2.1 Cell lines

The human embryonic kidney cell line expressing SV40 large T antigen (HEK293T) was provided by Palmer's lab at Stanford University (Stanford, CA). U87MG, a human glioblastoma cell line, and NK92 were purchased from Procell Biotechnology (Wuhan, China), while U87 expressing luciferase, U87<sup>LUCI</sup>, and HT1080, a human colon cancer cell line, were purchased from TongPai Biotechnology Co., Ltd (Shanghai, China). U87 constantly expressing mCherry, U87<sup>mCherry</sup> was generated in our lab.

U87MG, U87<sup>LUCI</sup>, and U87<sup>mCherry</sup> were cultured in MEM (Procell Biotechnology, China) media supplemented with 2 mM L-glutamine (Gibco, USA) and 10% heat-inactivated FBS (Gibco, USA). In addition, HEK293T and HT1080 were maintained in DMEM (Gibco, USA) media supplemented with 2 mM L-glutamine (Gibco, USA), 1% non-essential amino acids (Gibco, USA), and 10% heat-inactivated FBS. PBMC-derived iNSC cell line was generated and reversed in our lab and cultured as described previously [24]. NK92 cell line was cultured in X-VIVO-15 medium (Lonza, USA) supplemented with 10% heat-inactivated FBS.

### 2.2 GD2-specific CAR design and virus production

GD2-specific chimeric antigen receptor (GD2CAR) was generated with proper optimization [34]. GD2-specific VL and VH encoding sequences were synthesized referring to a patent (CN103347894B), and the sequences were sub-cloned into the template (pLenti6.4-CMV-CD19 hsCAR-T) to replace CD19-specific VL and VH to generate a GD2-specific CAR expression vector. The EGFP encoding sequence was amplified from the plasmid (pEGFP-N2) and fused with GD2-specific CAR by auto-cleaved peptide, P2A, as the reporter gene.

GD2CAR lentivirus was generated and stored at −80 °C. The viral titers were determined by a limiting dilution method on the HT1080 cell line using flow cytometry [35].

### 2.3 Generation of GD2-specific CAR-NK92

NK92 cell line was harvested in the logarithmic growth phase and adjusted to  $3 \times 10^6$ /mL by the complete medium. The suspension containing polybrene (10  $\mu$ g/mL) was mixed with virus stock at the multiplicities of infection (MOI) of 10. The mixture was seeded into the 24-wells plate coated with RetroNectin (10  $\mu$ g/mL) after incubation for 15 min. The treated NK92 was cultivated in the incubator, and after 6 h, the fresh medium was refilled. After 72 h, EGFP-positive cells were isolated and harvested using BD FACSAria (BD, USA) for subsequent cultivation. The purified GD2NK92 was cultivated for future study.

### 2.4 Generation of TK-transduced iNSCs

The encoding sequence of HSV-TK was obtained from UniProtKB (<https://www.uniprot.org/uniprotkb/Q9QNF7>), and the cDNA was chemically synthesized and sub-cloned into pLenti6.4-CMV-GD2CAR-P2A-EGFP to replace GD2CAR for obtaining pLenti6.4-CMV-TK-P2A-EGFP expression vector. The Virus expressing TK fused with EGFP was generated [34], the titer was measured using the HT1080 cell line, and the stock was stored at  $-80^\circ\text{C}$ . The iNSC cell line was cultured as described previously to generate TK-transduced iNSCs (iNSCs<sup>TK</sup>) [24]. The cells were harvested in the logarithmic growth phase and resuspended by the complete medium (polybrene, 5  $\mu$ g/mL) at  $5 \times 10^6$ /mL density for virus infection. TK-positive iNSCs were purified using BD FACSAria and cultivated for future study.

### 2.5 Immunofluorescence staining

iNSCs and iNSCs<sup>TK</sup> were cultured in a NSC proliferation medium and seeded on the PDL/laminin-coated coverslips. After 48 h of incubation, cells were fixed with 4% paraformaldehyde (PFA) for detecting NSC-associated marker expression using mouse anti-human NESTIN antibody (BD Biosciences, USA) and goat anti-human SOX2 antibody (Santa Cruz, USA), respectively. iNSC<sup>TK</sup> were stained with rabbit anti-GFP antibodies (Genscript, China).

For tissue samples, mice brains were fixed with 4% PFA overnight at  $4^\circ\text{C}$ , then transferred to 30% sucrose solution for 48 h for dehydration. Subsequently, the dehydrated brain tissues were frozen and sectioned into slices (20  $\mu$ m) for immunofluorescent staining. The secondary antibodies were used following a ratio that included, Cy3-conjugated donkey anti-rabbit/goat/mouse (1:400), FITC-conjugated donkey anti-rabbit/goat/mouse (1:200), and Cy5-conjugated donkey anti-rabbit/mouse/goat/rat (1:200). All of the antibodies were purchased from Jackson ImmunoResearch, USA.

### 2.6 iNSCs<sup>TK</sup>-mediated cytotoxicity assay

A lactate dehydrogenase (LDH) release assay was used to evaluate iNSCs<sup>TK</sup>-mediated cytotoxicity. iNSCs<sup>TK</sup> were suspended in the proliferation medium and seeded into the non-adhesive plate. The GCV treatment group was maintained in the conditional medium containing GCV (Selleckchem, USA) at 25  $\mu$ M, 50  $\mu$ M, and 100  $\mu$ M, respectively. The supernatant was collected by centrifugation after 72 h of cultivation to detect the LDH level, according to the LDH kit instructions (Promega, USA).

### 2.7 *In vitro* tumor tropism of iNSCs<sup>TK</sup>

The iNSCs<sup>TK</sup> tumor-specific migratory capacity toward U87<sup>mCherry</sup> glioma cells was evaluated by using transwell plates (Millipore, Germany). In brief, U87<sup>mCherry</sup> cells (50,000/0.6 mL) were placed in the lower chamber of transwell plates and incubated in a serum-free medium for 24 h. Subsequently, iNSCs<sup>TK</sup> cells (20,000/0.2 mL) were suspended in the serum-free medium in the upper inserts. Serum-free medium was used as a negative control for trigger factors. The non-migratory cells were removed from the insert after incubation for 3, 4, and 6 h at  $37^\circ\text{C}$ , and the filters were fixed with 4% paraformaldehyde (PFA) following staining with the DAPI (Richard-Allan Scientific, USA) to quantify the migratory cells. All experiments were conducted in triplicate, and the nuclei were counted randomly in five fields and expressed as means  $\pm$  SE.

### 2.8 *In vivo* tumor-homing study of iNSCs<sup>TK</sup>

U87<sup>mCherry</sup> cells were injected into the right lateral ventricle ( $-2.0$  mm right and  $+0.5$  caudal from bregma) at a depth of 3.5 mm to determine whether iNSCs<sup>TK</sup> was tropic toward glioma *in vivo*. iNSCs<sup>TK</sup> were stereotactically implanted into the right prefrontal cortex (1.5 mm upper to the tumor) at a rate of 0.5  $\mu$ L/min at coordinates after five days. Subsequently, 3 days later, mice were sacrificed and brain tissues collected. Frozen brain sections were prepared (20  $\mu$ m) and stained with anti-mCherry antibody (Invitrogen, USA) and anti-GFP antibody (Genscript, China) to examine the presence of the tumor cells and iNSCs<sup>TK</sup>, respectively.

### 2.9 *In vitro* anti-tumor bystander effect of iNSCs<sup>TK</sup>

iNSCs<sup>TK</sup> was co-transplanted with U87<sup>mCherry</sup> or U87<sup>LUCI</sup> cells at different ratios into PDL/laminin-coated cell culture plates. The medium was replaced with the conditional medium containing GCV (50  $\mu$ M) after 24 h. The bystander effect of iNSCs<sup>TK</sup>/GCV system on U87<sup>mCherry</sup> or U87<sup>LUCI</sup> cells was detected using confocal microscopy or an IVIS Lumina K Serial III System (Caliper, USA) after 72 h,

respectively. U87<sup>mCherry</sup> cells were counted in five fields and expressed as means  $\pm$  SE. The luciferase activity of U87<sup>LUCI</sup> was calculated using Living Image Software (Caliper, USA).

### 2.10 *In vitro* GD2NK92-mediated cytotoxicity assay

The cytotoxic activity on U87<sup>LUCI</sup> was conducted using the standard LDH release method. U87<sup>LUCI</sup> cells were incubated with GD2NK92 or NK92 at various effector/target (E/T) ratios (8:1, 4:1, and 2:1) in non-adhesive 96-well plates. After 5 h of cultivation at 37 °C, the supernatant was harvested and used for cytotoxicity measurement using CytoTox 96 Non-Radioactive Cytotoxicity Assay Kit following the manufacturer's instructions.

### 2.11 Cytokines assay

NK-mediated cytotoxic cytokines (IFN- $\gamma$  and TNF- $\alpha$ ) were tested by ELISA assay. Briefly, supernatant samples were collected and measured using ELISA Kits (Neobioscience, China), and the OD450 values were detected using VARIO SKAN FLASH (Thermo Fisher Scientific).

### 2.12 Animal study

Female NOD/SCID IL2R $\gamma$ c<sup>-/-</sup> NSG mice (6- to 8-week-old) were purchased from the Beijing Vitalstar Biotechnology Co., Ltd. To determine the activity of iNSCs<sup>TK</sup> bystander effect of killing U87 *in vivo*, NSG mice were anesthetized with chlorine ammonia ketone (100 mg/kg) and placed in a stereotaxic apparatus; U87<sup>LUCI</sup> cells ( $1.5 \times 10^5$ ) and iNSCs<sup>TK</sup> ( $4.5 \times 10^5$ ) were suspended in phosphate-buffered saline (PBS) and co-transplanted stereotactically into the right frontal lobe of mice via 10  $\mu$ L-Hamilton syringes. Mice in the control group were injected intraperitoneally with PBS (400  $\mu$ L) from day 1 to day 6. Animals in the GCV treatment group received intraperitoneally GCV (100 mg/kg) injections from day 1, once daily, for six days at 100 mg/kg. Mice in GCV + GD2NK92 treatment received intraperitoneal injection of GCV (similar to the GCV treatment group) followed by tail vein injection of  $2 \times 10^6$  GD2NK92 cells once per week for three weeks.

Tumor progression was evaluated at indicated time points by serial bioluminescence imaging. Mice were anesthetized with chlorine ammonia ketone (100 mg/kg) and then injected intraperitoneally with D-luciferin substrate (Promega, USA) resuspended in PBS (15 mg/kg). Imaging using an IVIS Lumina K Serial III System (Caliper, USA) was performed after 10 min of D-luciferin injection in small binning mode with an acquisition time rank from 1 s to 1 min to obtain unsaturated images. The animal experiment ethics committee of the capital medical university Xuanwu hospital (Beijing, China) approved the animal studies.

### 2.13 Statistical analysis

Student's t-test was applied to compare data between the two groups. One-way analysis of variance (ANOVA), two-way ANOVA, and repeated measures were applied for comparison among data sets from more than two groups. Survival among different treatment groups was compared using an exact log-rank test. Data were represented as mean  $\pm$  S.E.M., and differences were considered significant at  $P < 0.05$ .

## 3 Results

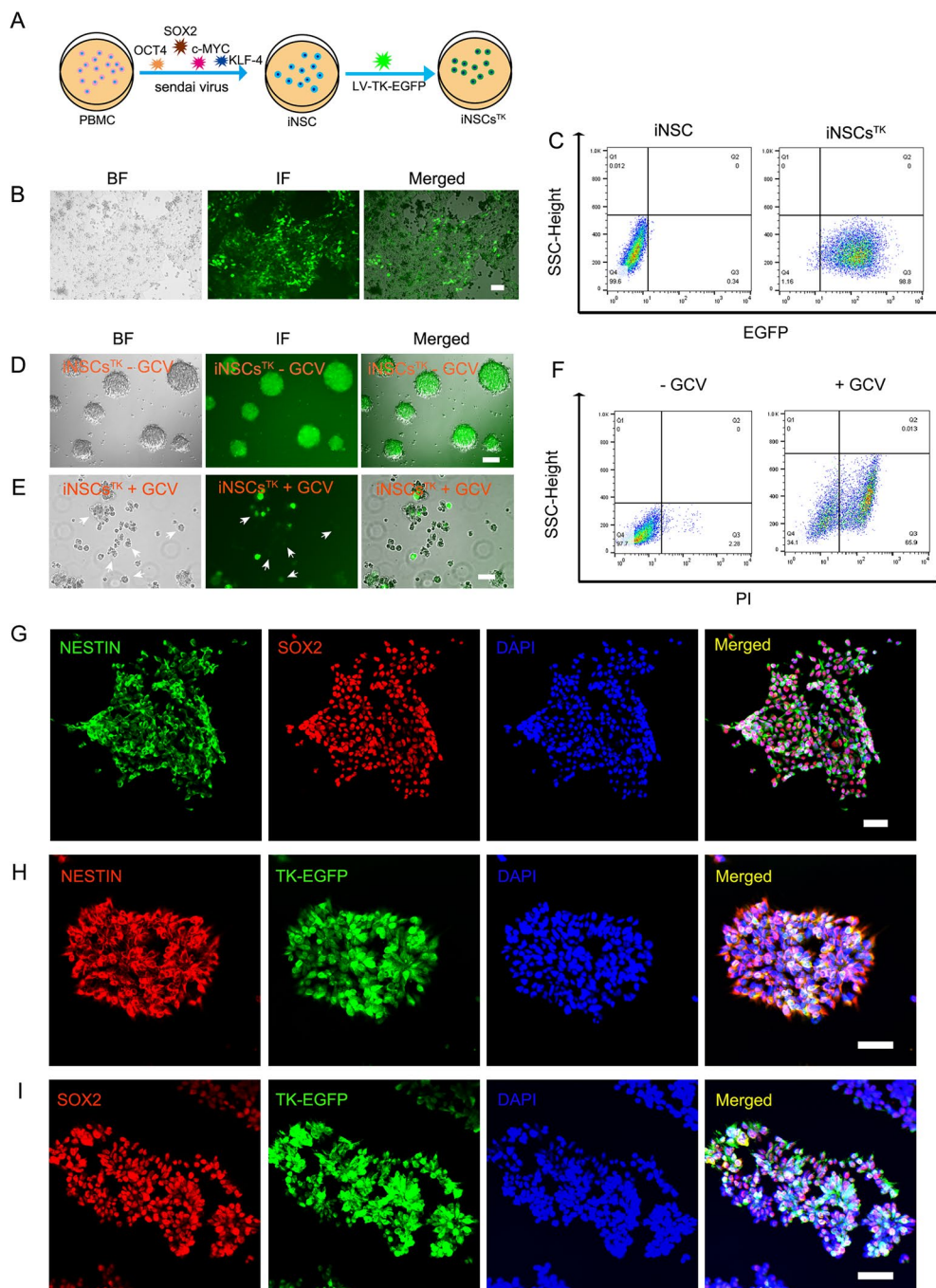
### 3.1 Generation and identification of therapeutic iNSCs expressing HSV-TK fused with EGFP

The preparation of therapeutic iNSCs expressing HSV-TK fused with EGFP was illustrated in Fig. 1A. Briefly, PBMC-derived iNSCs were dissociated, resuspended in the iNSCs proliferation medium, and transplanted into PDL/laminin-coated 24-well cell culture plate. After overnight cultivation, iNSCs were infected with lentivirus particles expressing HSV-TK fused with EGFP, diluted by infection medium (5  $\mu$ g/mL polybrene) at the MOI of 10. As shown in Fig. 1B, iNSCs could be infected effectively by lentivirus. Flow cytometry displayed that over 98% of the population was EGFP positive, indicating that HSV-TK was transduced successfully into iNSCs for effective expression (Fig. 1C).

The TK-fused EGFP-positive iNSCs were cultured in an NSC proliferation medium with or without GCV (25  $\mu$ M) for 72 h to determine the functional thymidine kinase (TK) expression in iNSCs. The results showed that TK-EGFP-positive iNSCs in the control medium could proliferate normally and form neurospheres (Fig. 1D). GCV-treated TK-EGFP positive iNSCs were significantly undergoing growth inhibition and triggered autolysis (Fig. 1E). Flow cytometric assay confirmed that considerable GCV-treated TK-EGFP positive iNSCs displayed greater value in the SSC-Height voltage gate. Moreover, the majority of this subpopulation was Propidium iodide (PI)-positive compared with the control group (Fig. 1F). In addition, iNSCs expressing only EGFP could proliferate normally and form neurospheres in the conditional medium containing 25  $\mu$ M GCV, the same as when cultured in the regular medium, suggesting that GCV was not cytotoxic to iNSCs without HSV-TK expression (Fig. S1). Overall, these results indicated that the HSV-TK gene was stable and functionally expressed in iNSCs.

The expression of several standard markers of iNSCs was detected via immunofluorescence staining to investigate iNSCs expressing HSV-TK whether maintaining the functional characteristics of NSC. iNSCs<sup>TK</sup> were found positive for NESTIN and SOX2 as their parental cells





**Fig. 1** Generation and identification of therapeutic iNSCs expressing HSV-TK (iNSCs<sup>TK</sup>). **A** Schematic diagram of generating therapeutic iNSCs<sup>TK</sup>. **B** PBMC-derived iNSCs expressing TK-P2A-EGFP were cultured in an NSC proliferation medium with G418. **C** Flow cytometry indicated that >98% of transfected iNSCs were EGFP<sup>+</sup>. **E** and **D** GCV effectively induced iNSCs<sup>TK</sup> autolysis at 25  $\mu$ M after cultivation

for 72 h. **F** Flow cytometry indicated that the percentage of PI-positive iNSCs<sup>TK</sup> cells increased significantly when cultured in a conditional medium with GCV (25  $\mu$ M) for 72 h. **G** Through immunofluorescent staining, NSC-associated markers, NESTIN and SOX2 could be effectively detected in iNSCs. **(H and I)** Therapeutic iNSCs<sup>TK</sup> cells were positive for NESTIN and SOX2. The scale bar was 50  $\mu$ m

(Fig. 1G, H, and I). Together, these results suggested that TK fused with EGFP was stable and effectively transduced and expressed in iNSCs, and this fusion protein was functional to trigger host cell autolysis when treated

by a specific substrate, GCV. Moreover, over-expression of HSV-TK would not interrupt the expression of iNSCs biological markers. Thus, therapeutic iNSCs<sup>TK</sup> was generated successfully.

### 3.2 *In vitro* and *in vivo* tumor tropism of iNSCs<sup>TK</sup>

Neural stem cells (NSC) harbor the capacity of extensive tropism toward glioblastoma [10]. The tropism level of iNSCs<sup>TK</sup> towards glioblastoma *in vitro* was tested by migration assay by using the transwell plates (Fig. 2A). Briefly, U87<sup>mCherry</sup> cells suspended in 0.6 mL serum-free medium were seeded in the lower chambers of the 24-transwell culture plates following 24 h cultivation. Subsequently, iNSCs<sup>TK</sup> suspended in 0.2 mL serum-free medium were seeded into the upper inserts. iNSCs<sup>TK</sup> and U87<sup>mCherry</sup> cells were co-cultured for 3, 4, and 6 h at 37 °C, respectively. Serum-free medium served as the negative control for trigger factors. Figure 2B and C indicate that iNSCs<sup>TK</sup> significantly migrated toward U87<sup>mCherry</sup> cells compared with the corresponding control group, consistent with other reports [18, 19].

The parallel tumor-specific tropism *in vivo* was investigated using an orthotopic xenograft tumor model of NSG mice, in order to assess the therapeutic effect of iNSCs<sup>TK</sup>. Briefly, U87<sup>mCherry</sup> cells were injected into the mouse brains at the right lateral ventricle. iNSCs<sup>TK</sup> were stereotactically implanted into the right prefrontal cortex (1.5 mm upper to the tumor) 5 days later at a rate of 0.5  $\mu$ L/min with the set coordinates. Mice were sacrificed 3 days following iNSCs<sup>TK</sup> transplantation, and the brains were collected to prepare frozen tissue sections (20  $\mu$ m), and then stained with anti-mCherry antibody and anti-GFP antibody to identify the tumor cells and iNSCs<sup>TK</sup> cells, respectively. The staining results showed that a mass of the implanted iNSCs<sup>TK</sup> had migrated toward the tumor tissue margin (Fig. 2D, upper panel), and some of the iNSCs<sup>TK</sup> had even migrated into the tumor bed (Fig. 2E), indicating that iNSCs<sup>TK</sup> displayed considerable tumor-specific tropism migration consistent with Bagó et al. [19, 36].

Mouse brain cryosections were stained with antibodies against NESTIN and SOX2 to determine whether iNSCs<sup>TK</sup> surrounding the tumor microenvironment retained the typical biological properties of NSCs. The results showed that NESTIN and SOX2 were expressed commonly in iNSCs<sup>TK</sup> that had migrated into tumor tissue (Fig. S2). Together, PBMC-derived iNSCs showed significant *in vitro* and *in vivo* tumor-homing activity, with genetic modification with HSV-TK. Moreover, the therapeutic iNSCs<sup>TK</sup> surrounding the tumor microenvironment retained the ability of NSC marker expression.

*In vitro* anti-tumor activity against GBM via bystander effect mediated by iNSCs<sup>TK</sup> iNSCs<sup>TK</sup> underwent a significant level of autolysis when maintained in the conditional medium with 25  $\mu$ M GCV as described above. iNSCs<sup>TK</sup> were cultured in the medium containing different concentrations of GCV for 72 h, respectively, to find an optimal concentration of GCV for the subsequent studies. As shown

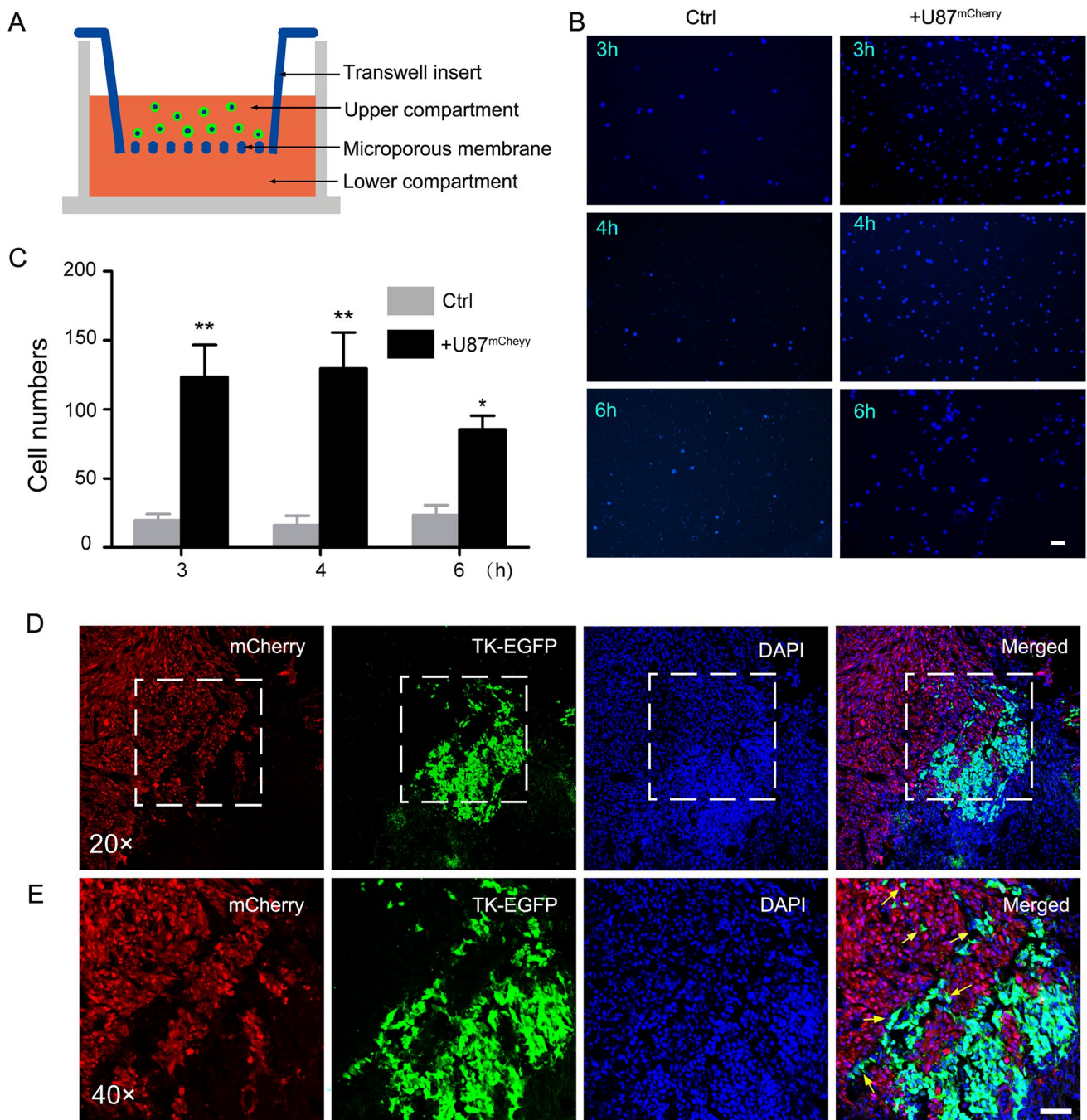
in Fig. 3A, GCV could efficaciously induce autolysis of iNSCs<sup>TK</sup> at all concentrations (25  $\mu$ M, 50  $\mu$ M, and 100  $\mu$ M). The highest level of released LDH was observed at 50  $\mu$ M GCV, which was decreased when the concentration increased to 100  $\mu$ M. Thus, 50  $\mu$ M GCV was the optimal concentration for iNSCs<sup>TK</sup> treatment, and 100  $\mu$ M GCV probably caused more cytotoxicity than TK-mediated autolysis (Fig. 3B). In addition, whether 50  $\mu$ M GCV would affect the viability of iNSC cells without TK expression was examined. Briefly, iNSC cells expressing EGFP only were dissociated into single cells and cultured in regular medium and conditional medium containing 50  $\mu$ M GCV, respectively. The results showed that 50  $\mu$ M GCV did not affect the proliferation or growth of iNSCs without TK expression (Fig. S3). Therefore, 50  $\mu$ M was used in the subsequent studies. iNSCs<sup>TK</sup> and U87<sup>LUCI</sup> cells were mixed in the ratios 0:1, 1:1, 2:1, 3:1, 4:1, and 5:1, respectively, and then co-cultured as described above. IVIS Lumina living images indicated that iNSCs<sup>TK</sup> showed evident anti-tumor activity against target cells at all ratios, and the cytotoxic efficiency was enhanced with increasing E/T ratio (Fig. 3C and D). However, there was no obvious impact on the viability of U87<sup>LUCI</sup> in the absence of GCV in the co-culture system (Fig. 3E and F). Therefore, iNSCs<sup>TK</sup> could mediate the effective cytotoxic activity on tumor cells at 50  $\mu$ M GCV.

U87<sup>mCherry</sup> cells were also co-cultured with iNSCs<sup>TK</sup> at the ratio of 1:3 to investigate the anti-tumor bystander effect of iNSCs<sup>TK</sup>. The results were consistent with those observed in the assay using U87<sup>LUCI</sup> cells. iNSCs<sup>TK</sup> could induce U87<sup>mCherry</sup> cell lysis through the bystander cytotoxicity effect in the presence of 50  $\mu$ M GCV (Fig. 3G–I). Besides, the iNSCs expressing EGFP only did not affect the viability of U87<sup>mCherry</sup> in the co-culture system with or without GCV (Fig. S4). It was remarkable that GBM cell line proliferated and grew normally in the conditional medium containing the same concentration of GCV (Fig. S5). These results suggested that GCV at the tested concentrations had no TK-independent cytotoxicity to iNSCs or tumor cells. Overall, the therapeutic iNSCs<sup>TK</sup> cells exhibited efficacious *in vitro* anti-tumor bystander cytotoxicity effect against GBM by cell–cell connection. However, neither U87<sup>LUCI</sup> nor U87<sup>mCherry</sup> was eliminated fully by iNSCs<sup>TK</sup> with GCV through the increased E/T ratio, indicating that some tumor cells were not sensitive to TK/GCV system. Thus, a combined therapeutics strategy would be a more effective choice.

### 3.3 GD2NK92-mediated *in vitro* cytotoxicity against GBM

Chimeric antigen receptor (CAR) engineered T/NK cells are considered a promising biological candidate for anti-tumor therapy against hematological and solid malignancies, including GBM [28, 31, 34]. Multiple studies



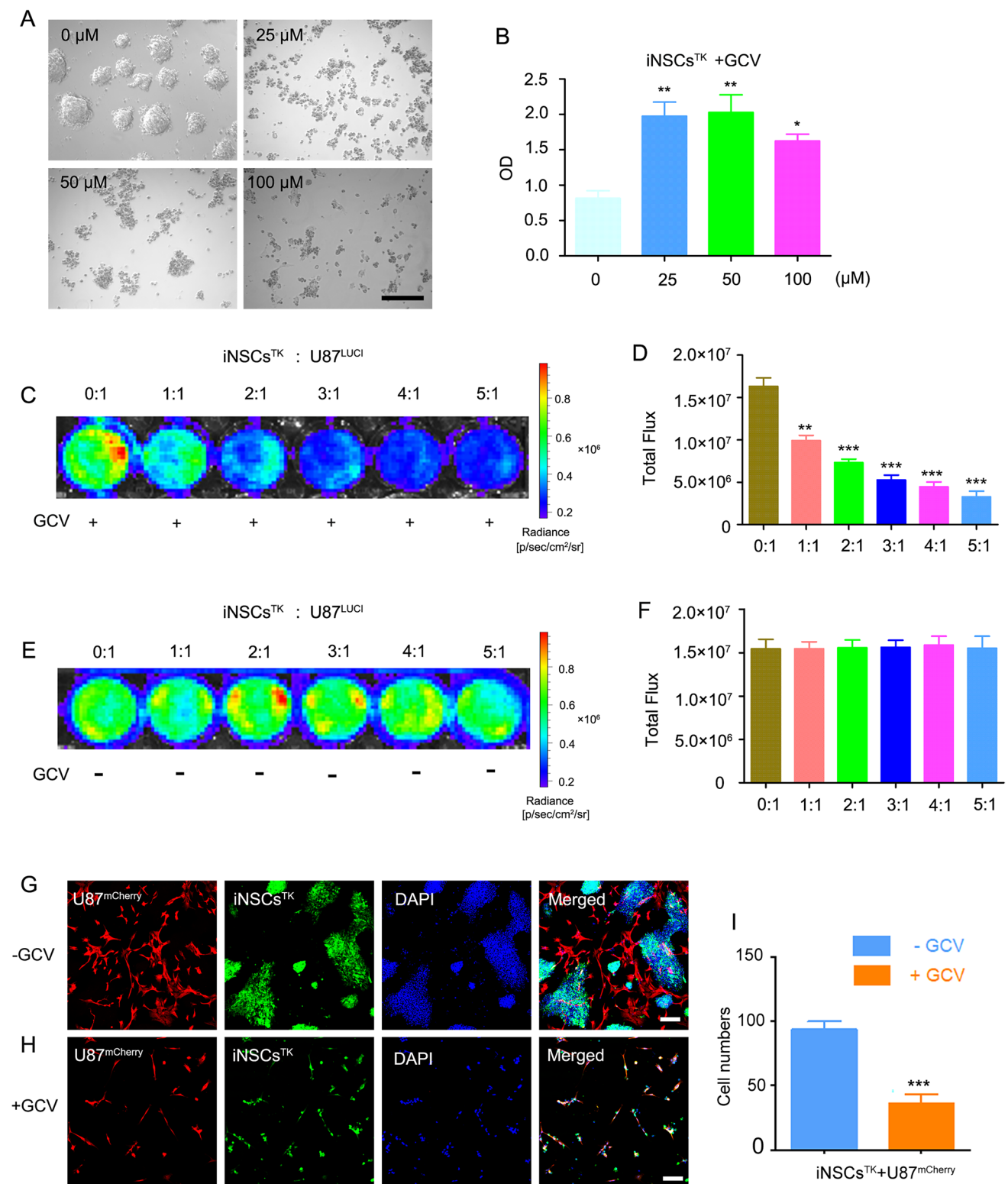


**Fig. 2** Therapeutic iNSCs<sup>TK</sup> migrate toward U87<sup>mCherry</sup> cells *in vitro* and *in vivo*. **A** Schematic diagram of the migration assay via a transwell system *in vitro*. **B** U87<sup>mCherry</sup> cells were cultured in the lower chamber with serum-free medium for 24 h. Subsequently, iNSCs<sup>TK</sup> were seeded into the insert in a serum-free medium. Transwell plates were harvested after cultivation for 3, 4, and 6 h, respectively. Serum-

free medium was served as the negative control. **C** The migratory iNSCs<sup>TK</sup> was counted randomly in five fields for each group. Three independent experiments were conducted. Significance levels were representative as \*  $P < 0.05$  and \*\* $P < 0.01$ . **D** and **E** Therapeutic iNSCs<sup>TK</sup> migrated toward to the engrafted tumor in NSG mice and invaded the tumor bed (yellow arrows). The scale bar was 50 μm

have demonstrated that GBM-associated antigen-specific CAR-T/NK displayed considerable pre-clinical benefits in the treatment of glioblastoma [28, 29, 37]. GD2 is a tissue-specific marker of glioblastoma and is considered an ideal target for CAR-based immune cellular therapy

[29, 30, 38]. Herein, a second-generation CAR that specifically recognized and bonded with the GD2 expressed on the tumor cell surface was designed to eliminate tumor cells that were not sensitive to iNSCs<sup>TK</sup>/GCV system (Fig. 4A).



Flow cytometric assay showed that ~30% U87<sup>LUCi</sup> cells were GD2 positive, indicating that not all U87 cells could be eliminated by GD2-specific CAR-mediated cytotoxicity but could be used as supplementary therapeutics for iNSCs<sup>TK</sup>/GCV system (Fig. 4B). A gene encoding GD2-specific CAR

fused with EGFP was transduced into NK92 via lentivirus to generate GD2-specific CAR-engineered NK92 (GD2NK92). Subsequently, the EGFP-positive NK92 cells were harvested by fluorescence-activated cell sorting (FACS). The flow cytometric assay indicated that > 70% of NK92 were



**Fig. 3** iNSCs<sup>TK</sup> cells exhibit evident *in vitro* bystander anti-tumor effect against GBM cells. **A** iNSCs<sup>TK</sup> cells were cultured in a conditional medium containing 0, 25, 50, and 100  $\mu$ M of GCV for 72 h, respectively. **B** LDH release assay for cytotoxicity induced by iNSCs<sup>TK</sup> in the presence of GCV. **C** and **E** Representative images of U87<sup>LUCI</sup> activity in GCV-treated and without GCV groups. U87<sup>LUCI</sup> cells and iNSCs<sup>TK</sup> were seeded onto PDL/laminin-coated cell culture plates. Co-culture medium was replaced by a conditioning medium containing 50  $\mu$ M GCV or no GCV after 24 h, and the plates were detected by IVIS Lumina living image system following the 72-h co-culture. **D** and **F** The total luciferase activity was detected in at least three independent replicates in GCV-treated and control groups. **H**, **G** and **I** Confocal images of iNSCs<sup>TK</sup> and U87<sup>mCherry</sup> co-culture with GCV or without GCV cells were managed as described above. Cell numbers of U87<sup>mCherry</sup> were counted in five random fields in each group; three independent experiments were conducted. Significance was represented as \*  $P < 0.05$ , \*\*  $P < 0.01$ , \*\*\*  $P < 0.001$ . The scale bar was 100  $\mu$ m

EGFP+, indicating that GD2-specific CAR could be widely expressed in the NK92 cell lines (Fig. 4C). U87<sup>LUCI</sup> cells were co-cultured with NK92 or GD2NK92 at various ratios of effector to target (E/T) indicated as 8:1, 4:1, and 2:1, respectively, to investigate the anti-tumor activity exerted by GD2NK92. The results showed that GD2NK92 cells displayed a more robust anti-tumor effect than NK92 cells, particularly at the ratio of 8:1 (Fig. 4D).

Proinflammatory cytokines secreted by NK92 and GD2NK92 were detected *in vitro* to explore the potential mechanisms by which GD2NK92 had a stronger anti-tumor cytotoxicity than NK92. IFN- $\gamma$  and TNF- $\alpha$  derived from the culture supernatant were examined by using ELISA. Figure 4E showed that GD2NK92 had a stronger capacity to secrete IFN- $\gamma$  than NK92 in the regular cultivation condition. Although secreted IFN- $\gamma$  levels of NK92 and GD2NK92, when co-cultured with U87<sup>LUCI</sup>, significantly increased compared with the corresponding control (Fig. 4F and G), GD2NK92 displayed a more considerable level of IFN- $\gamma$  secretion than that of NK92 (Fig. 4H). The ability to secrete IFN- $\gamma$ , but not TNF- $\alpha$ , was significantly enhanced in NK92 cells engineered with GD2-specific CAR containing 4-1BB co-stimulator domain and CD3 $\zeta$  stimulation domain either in the regular cultivation or co-cultured with U87<sup>LUCI</sup> (Fig. 4I, J and K).

The active and inhibitory receptors on NK92 and GD2NK92 were also assessed to investigate the potential mechanisms associated with the enhancement of anti-tumor activity. Nkp30 was generally expressed on NK92 and GD2NK92 (Fig. 4M), however, about 50% of NK92 and GD2NK92 expressed Nkp44 (Fig. 4N). Nkp30 and Nkp44 expressed on GD2NK92 cellular surface were marginally higher than on NK92, with no significant difference between them. Likewise, CD94 was widely expressed on NK92 and GD2NK92, and its expression was slightly higher on GD2NK92 than on NK92 (Fig. 4O). TIM-3 expression on GD2NK92 and NK92 were at a comparatively low

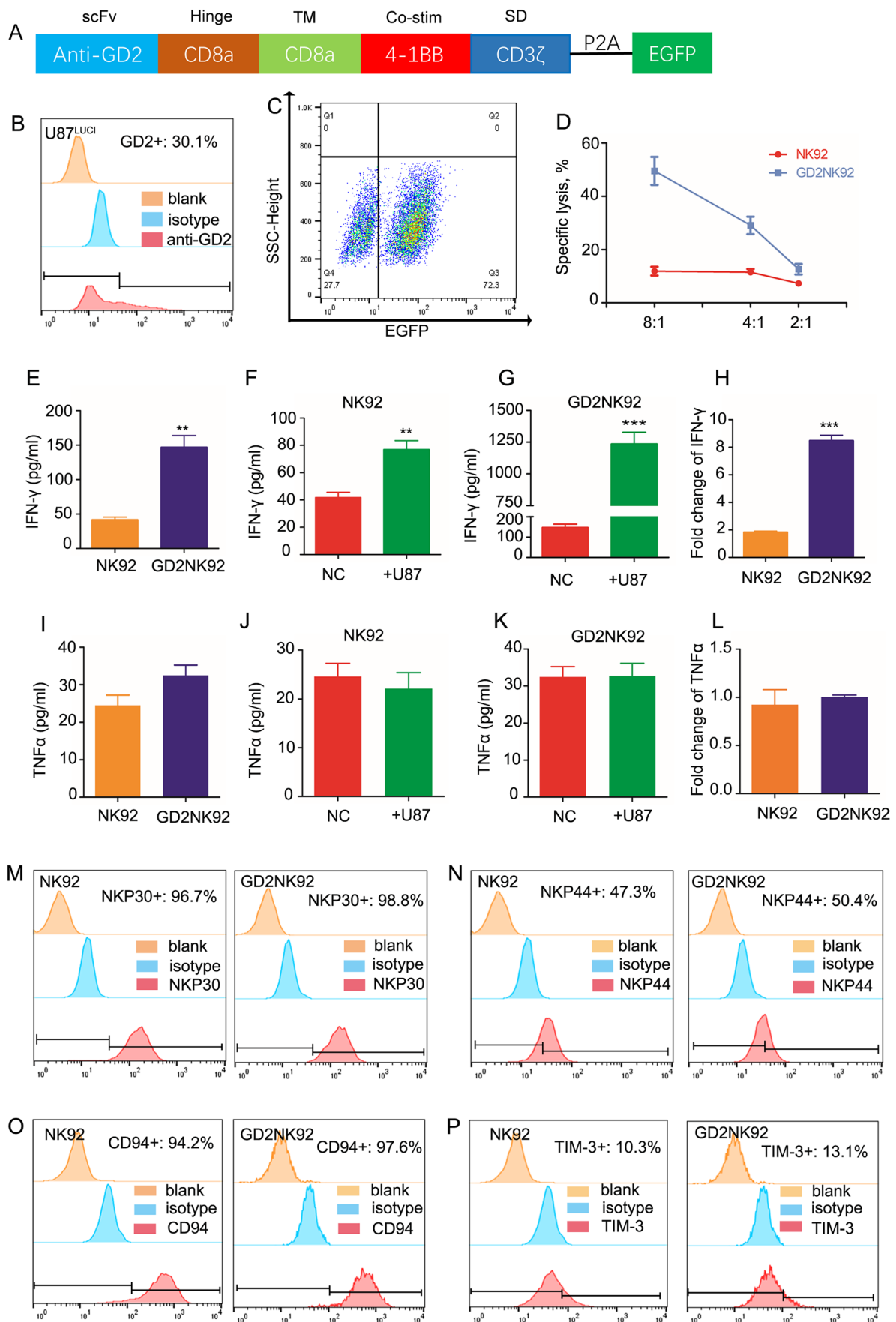
level—13.1% and 10.3%, respectively (Fig. 4P). Together, IFN- $\gamma$  secretion and anti-tumor activity were effectively enhanced by the 2<sup>nd</sup> generation CAR in NK92, which might be independent of the activation and inhibition receptor expression of NK cells.

### 3.4 GD2NK92 exhibit *in vitro* cytotoxicity on GBM cells not sensitive to iNSCs<sup>TK</sup>/GCV

Although the iNSCs<sup>TK</sup> exhibited an effective bystander anti-tumor effect on inducing autolysis of U87 cells in the presence of GCV, it was observed that some tumor cells were resistant to cytotoxic metabolites. Moreover, GD2NK92 cells could not kill the glioblastoma cells completely *in vitro* at the indicated E/T ratio (up to 1:8). Thus, the following experiments were designed to investigate whether the combination of iNSCs<sup>TK</sup>/GCV and GD2NK92 could eradicate the glioblastoma cells *in vitro* (Fig. 5A). U87<sup>LUCI</sup> and iNSCs<sup>TK</sup> were co-seeded onto PDL/laminin-coated 24-well culture plate at the ratio of 1:3, and U87<sup>LUCI</sup> cells alone served as control. The medium was replaced with a conditioned medium with 50  $\mu$ M GCV after 24 h of inoculation. Subsequently, cells were co-cultured for 72 h following GD2NK92 treatments for an additional 5 h. GD2NK92 was found to eliminate the residual tumor cells in the combination treatment group, which were eliminated more thoroughly by increasing the E/T ratios (Fig. 5B and C). All tumor cells receiving an E/T ratio of 8:1 GD2NK92 following iNSC<sup>TK</sup>/GCV treatment were dead, though few luciferase activities in the culture supernatant were found. Thus, GD2NK92 exhibited cytotoxicity on GCV-resistant glioblastoma cells. Additionally, whether the iNSCs<sup>TK</sup>/GCV system was cytotoxic to NK92 cells *in vitro* was examined. In brief, iNSCs<sup>TK</sup> cells were cultured in normal medium and conditional medium containing 50  $\mu$ M GCV for 72 h, respectively, and then GD2NK92 cells were added to the culture system for another 5 h. Cells were collected and labeled with anti-CD56 antibody and PI, and then the viability of GD2NK92 cells was examined by using flow cytometry. The results showed that iNSCs<sup>TK</sup>/GCV system did not affect the viability of GD2NK92 cells (Fig. S6). Together, combination therapy of iNSCs<sup>TK</sup>/GCV and GD2NK92 against GBM showed a superior anti-tumor effect.

### 3.5 Therapeutic anti-tumor effect mediated by combination of iNSCs<sup>TK</sup>/GCV and GD2NK92 *in vivo*

The NSG mice tumor model was used in this study to investigate further the anti-tumor effect of the combination of iNSCs<sup>TK</sup> and GD2NK92 against glioblastoma *in vivo* (Fig. 6A). The results showed that iNSCs<sup>TK</sup>/GCV system exhibited an efficacious *in vivo* anti-tumor effect



**Fig. 4** *In vitro* cytotoxicity of GD2NK92 cells against GBM. **A** Schematic diagram of the anti-GD2 CAR structure. **B** GD2 expression on the U87<sup>LUCI</sup> cell surface was examined by using flow cytometry. **C** Detection of positive GD2NK92 cells by flow cytometry. **D** Cytotoxicity-mediated by GD2NK92 and NK92 against U87<sup>LUCI</sup>. **E** IFN- $\gamma$  secretion of NK92 and GD2NK92 in the regular cultivation condition detected by using ELISA. **F** and **G** Secretion of IFN- $\gamma$  increased significantly when stimulated by U87<sup>LUCI</sup> for 5 h both in NK92 and GD2NK92. **H** GD2NK92 cells exhibited a higher capacity of IFN- $\gamma$  secretion than NK92 cells when co-cultured with U87<sup>LUCI</sup> cells. **I–L** U87<sup>LUCI</sup> could induce neither NK92 nor GD2NK92 cells to secrete TNF- $\alpha$  in the co-culture system. **M–P** Detection of the active and inhibitory receptors expressed on NK92 cells and GD2NK92 cells by using flow cytometry. Significance was represented as \*\*  $P < 0.01$ , \*\*\*  $P < 0.001$

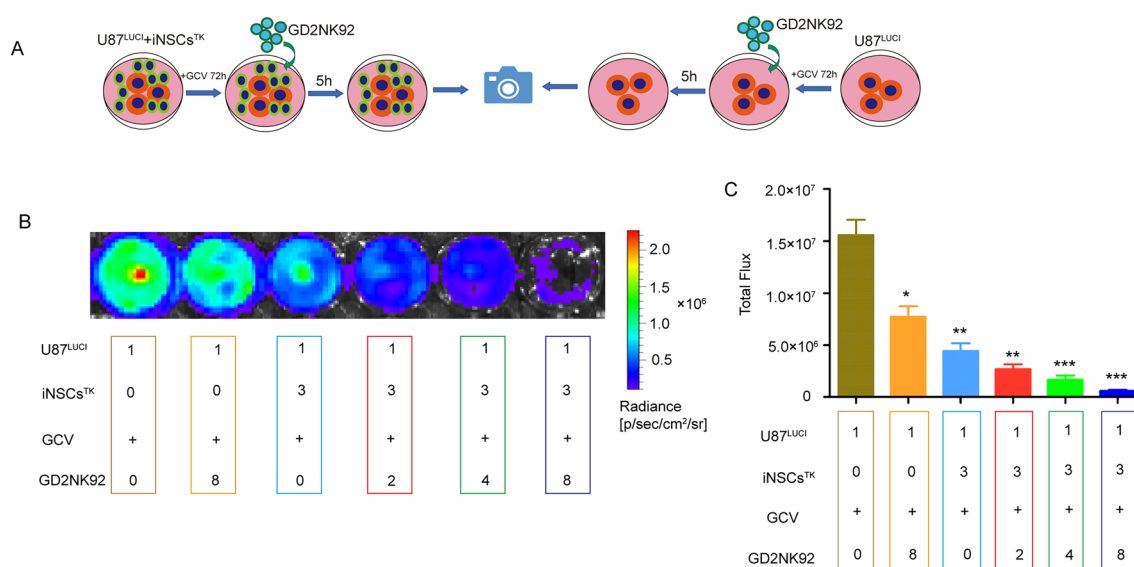
at the various indicated time points (Fig. 6B and C). The tumor burden was reduced, and the median survival time of tumor-bearing mice (PBS group: 28 days vs. GCV-treated group: 37 days) was significantly prolonged (Fig. 6D and E). Moreover, combination therapy of iNSCs<sup>TK</sup>/GCV and GD2NK92 showed a more potent anti-tumor activity in reducing tumor burden and prolonging the median survival time (GCV + GD2NK92 group: 48 days) in tumor-bearing mice (Fig. 6B–E).

The residual level iNSCs<sup>TK</sup> of in the brain of GCV-treated mice was evaluated to explore the potential

mechanisms associated with the increased anti-tumor efficiency of combined therapy *in vivo*. The results showed that GCV could completely induce autolysis of iNSCs<sup>TK</sup> in the tumor environment. However, there still remained some residual GBM cells (Fig. S7), consistent with the *in vitro* results that iNSCs<sup>TK</sup>/GCV could not eradicate all GBM cells.

Consequently, whether caspase-3 activation was involved in the anti-tumor effect induced by the combination therapeutic strategy was investigated. The frozen brain sections were prepared and stained with the antibody against active caspase-3. There was a significant level of caspase-3 activation in tumor-bearing mice treated with combined therapy (Fig. 6F). In contrast, activated caspase-3 was not detected in the tumor bed of PBS and GCV treated-mice (Fig. 6F). The enhanced tumor cell apoptosis might be a potential pathway associated with the combination therapy.

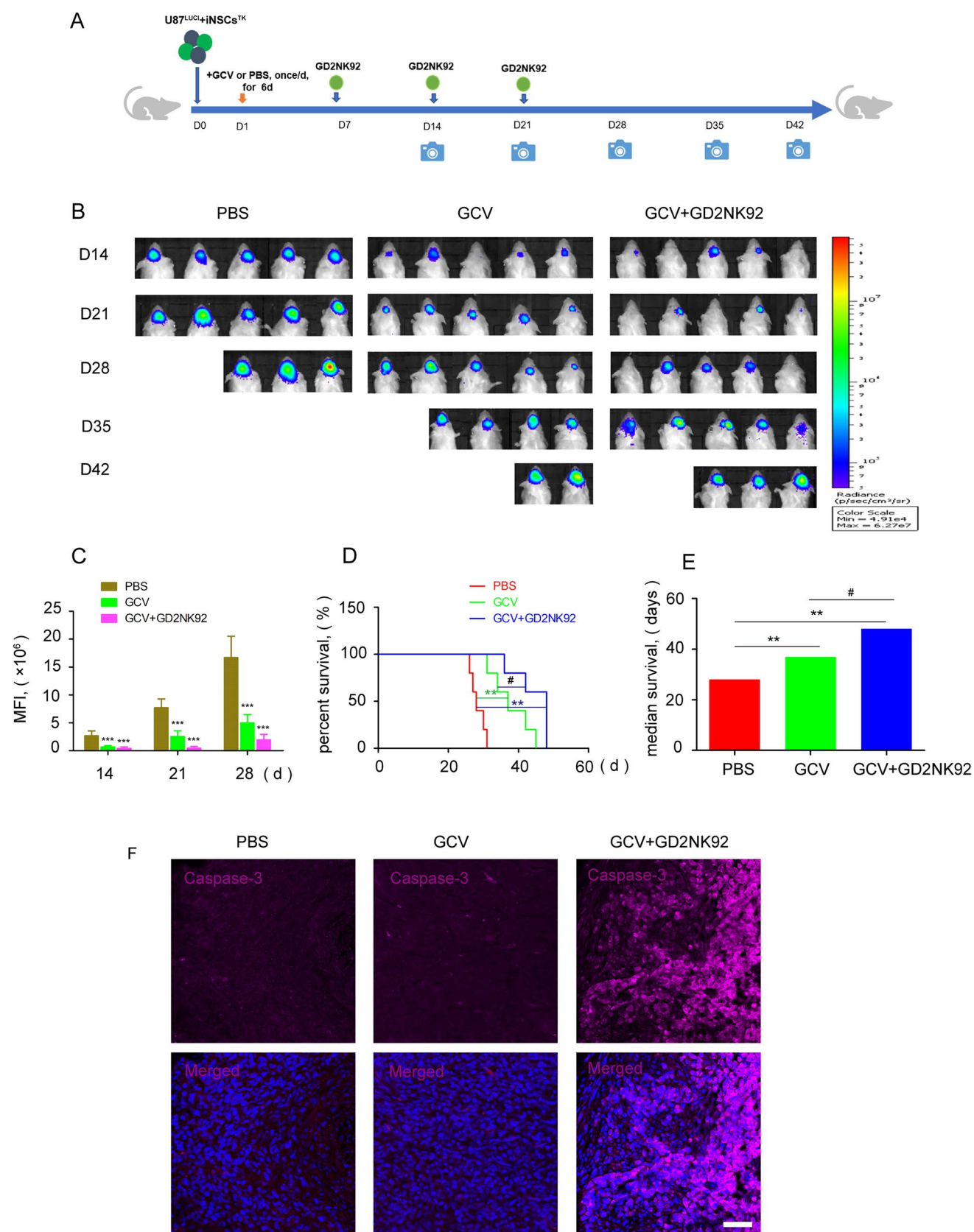
The anti-tumor capacity of GD2NK92 cells only *in vivo* was also investigated (Fig. S8). Briefly, U87<sup>LUCI</sup> cells ( $1.5 \times 10^5$ ) were transplanted stereotactically into the right frontal lobe of mice on day 0, and GD2NK92 cells ( $2 \times 10^6$ ) were injected through the tail vein on day 7, 14, and 21, respectively. GD2NK92 reduced the tumor burden in this model and prolonged the median survival time of mice (29 days vs. 36 days)



**Fig. 5** GD2NK92 cells exhibit *in vitro* cytotoxicity on TK/GCV-resistant glioblastoma cells. **A** Schematic diagram of the combinational therapy process. **B** U87<sup>LUCI</sup> cells were seeded or co-seeded with iNSCs<sup>TK</sup> onto PDL/laminin-coated cell culture plates. After 24 h, the culture medium was replaced with a conditional medium containing 50  $\mu$ M GCV and maintained for 72 h. The luciferase activity was examined after 5 h when the GD2NK92 cells were seeded

into the corresponding wells. GD2NK92 cells could further eliminate the residual tumor cells resistant to the iNSCs<sup>TK</sup>/GCV bystander-killing effect. **C** The total fluorescence intensity of each well was calculated, and luciferase activity was detected in at least three independent replicates. The numbers in (B) and (C) indicated the ratios of iNSCs<sup>TK</sup>/U87<sup>LUCI</sup> and GD2NK92/U87<sup>LUCI</sup>, respectively. Significance was represented as \*  $P < 0.05$ , \*\*  $P < 0.01$ , \*\*\*  $P < 0.001$





**Fig. 6** Combinational therapy of iNSCs<sup>TK</sup>/GCV and GD2NK92 cells against GBM *in vivo*. **A** Experimental schematic diagram of combinational therapy of iNSCs<sup>TK</sup>/GCV and GD2NK92 in xenograft mice model; U87<sup>LUC</sup> ( $1.5 \times 10^5$ ) and iNSCs<sup>TK</sup> ( $4.5 \times 10^5$ ) cells were co-transplanted into the right frontal lobe of NSG mice on day 0. Mice in the GCV-treated group intraperitoneally received an injection of GCV at a dosage of 100 mg/kg/d from day 1 to day 6; GCV+GD2NK92 group mice received an intravenous tail injection of GD2NK92 cells ( $2 \times 10^6$ ) which received the intraperitoneal injection of GCV as the GCV-treated mice, once a week, for three weeks; control group mice were intraperitoneally injected with PBS. **B** and **C**, iNSCs<sup>TK</sup>/GCV effectively suppressed the GBM progression compared with the PBS-treated mice. GD2NK92 cells could further reduce tumor burden in tumor-bearing mice which received intraperitoneal injection of GCV. **D** and **E**, iNSCs<sup>TK</sup>/GCV showed effectively anti-tumor activity for prolonging the median survival of tumor-bearing mice; the combinational therapy of iNSCs<sup>TK</sup>/GCV and GD2NK92 further improved the anti-tumor efficiency. **F** Caspase-3 activation was detected in the frozen mouse brain sections. \*\* $P < 0.01$ , \*\*\* $P < 0.001$ . The scale bar was 50  $\mu\text{m}$

(Fig. S8, B-D). However, few GBM cells were positive for caspase-3 in GD2NK92-treated mice (Fig. S8-F).

## 4 Discussion

Based on the results of pre-clinical studies, Phase I clinical trials were accomplished using cytotoxic NSCs as anti-GBM cell carriers [10, 14, 39]. In the clinical trials, application of cytosine deaminase (CD)-NSC has been proved as a safe and feasible therapeutic strategy, although the overall survival of GBM patients with recurrence was not improved [39]. However, ethical issues remain a challenge for using aborted fetal neural stem cells. In our study, adult human donor PBMC-derived iNSCs<sup>TK</sup> were used as efficient anti-tumor drug vehicles due to their tumor tropism. The iNSCs<sup>TK</sup>/GCV system exhibited evident bystander anti-tumor effect against GBM cells *in vitro*, reduced the GBM tumor burden and significantly prolonged the median survival time of tumor-bearing mice with therapeutic agents. These results proved patient- or donor-derived PBMCs as a feasible and safe alternative source for preparation of anti-GBM iNSCs, given the reduced risk of teratoma formation compared to iPSC-derived iNSCs [40].

GBM is a highly lethal brain tumor with a complex immunosuppressive microenvironment that contributes to the poor efficacy of immunotherapies that are otherwise successful in treatment of other immunogenic cancers [1, 25, 41]. Patient-derived primary glioblastoma cells frequently express GD2 antigen at a high level [29, 30]. In a recent phase I study, application of CAR-T cells was proved as a feasible and safe treatment for recurrent glioblastoma, demonstrating signs of clinical efficacy and transient responses in some patients [42]. However, another phase I trial using a

third-generation CAR construct of EGFRvIII failed to show therapeutic significance [43].

NK cells do not pose a high risk of inducing graft-vs-host-disease (GVHD) [31, 37, 44] and CAR-NK is generally considered an alternative promising therapeutic strategy for treating malignant tumors [44–46]. In this study, GD2-specific CAR-modified NK92 cells (GD2NK92) were successfully generated and exhibited a more potent anti-tumor cytotoxicity against GBM cells *in vitro* compared with that of parental NK92 cells. The cells were infused intravenously into xenografts models, which prolonged the median survival of tumor-bearing mice, suggesting that GD2-modified NK cell lines or GD2-modified primary NK cells may be promising candidates for the future development of immunotherapies for treatment of GBM.

Although both iNSCs<sup>TK</sup>/GCV system and GD2NK92 exhibited efficient anti-tumor effects against GBM, some residual tumor cells remained a challenge to be eradicated. The bystander effect of the HSV-TK system requires tight connections between cells, which might fail to induce autolysis of distant tumor cells that have not established cellular gap junctions [18, 47, 48]. Besides, certain GBM cells possess specific DNA damage repair mechanisms to prevent death and could be resistant to HSV-TK/GCV therapy [49, 50]. In addition, not all malignant cells express GD2 antigen, which might hamper the effect of GD2NK92. Therefore, combining the iNSCs<sup>TK</sup>/GCV system and GD2NK92 against GBM was investigated for the capacity to eliminate resistant GBM cells.

The combinational therapy showed that GD2NK92 cells could further eliminate the residual tumor cells following treatment with iNSCs<sup>TK</sup>/GCV *in vitro*, reduce the tumor burden and improve the median survival time in xenograft model mice. It is worth mentioning that mice were infused intravenously with GD2NK92 cells following iNSCs<sup>TK</sup>/GCV treatments in the present study. Thus, intravenous administration of GD2NK92 cells was probably capable to reach the intracranial tumor site post iNSCs<sup>TK</sup>/GCV treatments. For patients or physicians, intravenous infusion of GD2NK92 would be more economical and safer than intracranial injection, which requires complex surgical management.

Although the results of the current study were promising, there were some limitations. Firstly, the random insertion of lentivirus into genome may pose some risk, and the efficiency of lentiviral infection of NK cells remains low [51]. However, Yun Ge et al. developed an enzyme-mediated proximity cell labeling strategy which allows loading biomolecules onto the cell surface [52]. This strategy might provide more advantages for NK cell-based anti-tumor immunotherapy. Secondly, HSV-TK catalyzes non-toxic prodrug into cytotoxic metabolite triphospho-GCV and inhibits the DNA polymerase activity, leading to apoptosis of targeted tumor cells [8].

Furthermore, HMGB1 released from apoptotic cells might induce a potent immune response [53]. Thus, the risk of developing a local immune response after tail vein injection of GD2NK92 cells in tumor-bearing mice treated with GCV needs further investigation. Finally, the time interval of combination therapy between injections of iNSCs<sup>TK</sup>/GCV and GD2NK92 cells, and the optimal dosage and frequency of CAR-NK cell infusion should be further investigated.

## 5 Conclusion

In this study, we provided a pre-clinical proof of concept for a post-operative adjuvant therapy to treat GBM using a combinational cellular therapy. The combinational immunotherapeutic strategy was developed by using PBMC-derived iNSCs and a second-generation CAR-engineered NK cell line. iNSCs were genetically modified to express HSV-TK to eliminate tumor cells via a bystander effect, while GD2-specific CAR-NK92 could consequently exert CAR-mediated cytotoxicity targeting iNSCs<sup>TK</sup>/GCV resistant malignant cells, inducing a robust anti-tumor effect *in vitro* and in xenograft mouse models. The study offered a potential promising adjuvant treatment strategy against high-grade GBM.

**Abbreviations** GBM: Glioblastoma.; HSV-TK: Herpes simplex virus thymidine kinase.; CNS: Cancers of the central nervous system.; EGFP: Enhanced green fluorescent protein.; CAR: Chimeric antigen receptors.; iNSCs: Inducing neural stem cells.; LDH: Lactate dehydrogenase.; PI: Propidium iodide.; GD2: Disialoganglioside.; NK cell: Natural killing cell.; GCV: Ganciclovir.; HNA: Human nuclei antigen.; GVHD: Graft-vs-host-disease.; HMGB1: High mobility group box 1 protein

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**Author contributions** Z.C., W.L., Y.Z. and X.T. contributed to the conception and experimental designs; W.L., Y.Z., Z.L., G.Z., H.W., and X.Z. did the experiments and analyzed the data; W.L., Y.Z., Z.L., G.Z., H.W., and X.Z. collected the samples; W.L., Y.Z. and Z.C. wrote and revised the manuscript. Z.C. secured funding and supervised the study. All authors have read the manuscript and approved the final version.

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**Data availability** All data generated or analyzed during this study are included in this published article.

## Declarations

**Ethics approval and consent to participate** All Procedures performed in studies involving animals were approved by the Ethics Committee of Xuanwu hospital capital medical university. All the mouse experimental procedures were performed according to the protocols approved by

the Xuanwu hospital capital medical university Experimental Animal Care Commission.

**Consent for publication** All authors have agreed to publish this manuscript.

**Competing interests** The authors declare that they have no competing interests.

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